Simple Trigonometric Identities

Exam code: 9MA0

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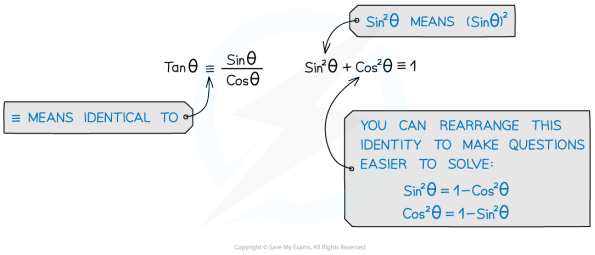
Help me understand

Simple trigonometric identities

What is a trigonometric identity?

- Trigonometric identities are statements that are true for all values of x or theta (θ)
- They are used to help simplify trig equations before solving them
- The first two identities you must know are:



Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Where do trigonometric identities come from?

FOR THE FIRST IDENTITY WE KNOW

$$\sin\theta = \frac{OPP}{HYP} \quad \text{AND} \quad \cos\theta = \frac{ADJ}{HYP} \quad \text{AND} \quad \tan\theta = \frac{OPP}{ADJ}$$

SO IF WE DIVIDE $\sin\theta$ BY $\cos\theta$

$$\frac{\sin\theta}{\cos\theta} = \frac{OPP \div HYP}{ADJ \div HYP} = \frac{OPP}{ADJ} = \tan\theta$$

SO

$$\tan\theta \equiv \frac{\sin\theta}{\cos\theta}$$

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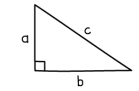


FOR THE SECOND IDENTITY WE NEED TO REMEMBER PYTHAGORAS' THEOREM

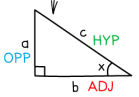
$$a^2 + b^2 = c^2$$

DIVIDING BY c^2

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{c^2}{c^2}$$

$$\left(\frac{a}{c}\right)^2 + \left(\frac{b}{c}\right)^2 = 1$$

IF WE RELABEL TRIANGLE FOR TRIG



$$\frac{a}{c} = \frac{OPP}{HYP} = \sin\theta \quad \text{AND} \quad \frac{b}{c} = \frac{ADJ}{HYP} = \cos\theta$$

SO $\sin^2\theta + \cos^2\theta \equiv 1$

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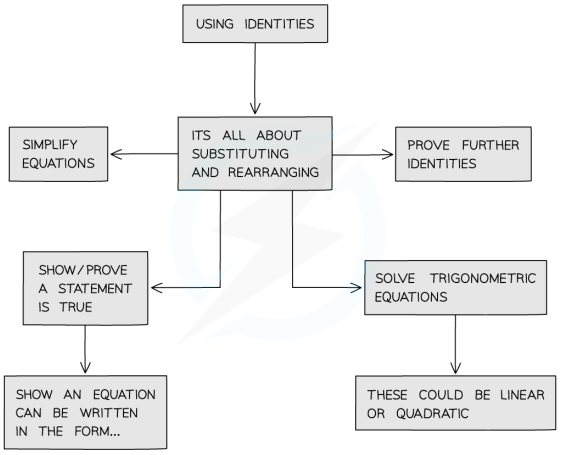
How do I use trigonometric identities?



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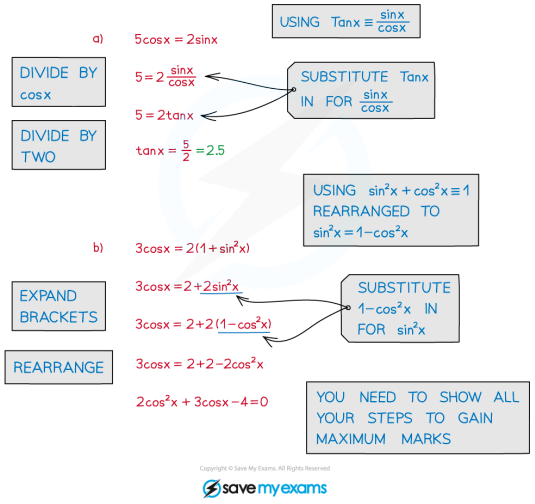
Examiner Tips and Tricks

- You'll need to remember these identities as they aren't in the formula booklet.

Worked Example

a) For an angle x, $5\cos x = 2\sin x$. Find $\tan x$.

b) Show that the equation $3\cos x = 2(1 + \sin^2 x)$ can be written in the form $2\cos^2 x + 3\cos x - 4 = 0$.



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